

# Plan

- Schur Unitary Triangularization Theorem
- Spectral Theorem
- Cayley-Hamilton Theorem

# Orthogonal and Unitary matrices

## Example

Consider the matrix  $A = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$ . Note that the columns of  $A$  are orthogonal. In fact, orthonormal. Check:  $AA^t = A^tA = I_2$ .

## Definition

- A **real** matrix  $A$  is said to be **orthogonal** if  $AA^t = I = A^tA$ .
- A matrix  $A$  is said to be **unitary** if  $AA^* = I = A^*A$ .

## Result

- A **real unitary** matrix is **orthogonal**.
- A real matrix is orthogonal **iff**  $A^{-1} = A^t$ .
- A matrix is unitary **iff**  $A^{-1} = A^*$ .
- If  $A$  is **orthogonal**, then  $A^t$  is orthogonal.
- If  $A$  is **unitary**, then  $A^*$  is unitary.

## Result

- An  $n \times n$  matrix  $A$  is *orthogonal / unitary* iff the columns and rows form *orthonormal bases* of  $\mathbb{R}^n / \mathbb{C}^n$ .

Suppose  $Q^t A Q = D = \text{diag}(\lambda_1, \dots, \lambda_n)$ , where  $Q$  is *orthogonal*. Then  $A \approx D$ ,  $\lambda_1, \dots, \lambda_n$  are (real) eigenvalues of  $A$ , and the columns of  $Q$  are corresponding eigenvectors.

## Definition

- A **real** matrix  $A$  is said to be *orthogonally diagonalizable* if there is an *orthogonal matrix*  $Q$  such that  $Q^t A Q$  is a *diagonal* matrix.
- A matrix  $A$  is said to be *unitarily diagonalizable* if there is an *unitary matrix*  $U$  such that  $U^* A U$  is a *diagonal* matrix.

## Result

A **real** matrix  $A$  is *orthogonally diagonalizable* iff all eigenvalues of  $A$  are *real* and there is an *orthonormal basis* of  $\mathbb{R}^n$  consisting of *eigenvectors* of  $A$ . (Similarly, for unitary.)

## Result (Schur Unitary Triangularization Theorem)

For an  $n \times n$  matrix  $A$ , *there is* an *unitary matrix*  $U$  such that  $U^*AU$  is upper triangular.

### Remark

Diagonal entries in  $U^*AU$  of the previous result are the *eigenvalues* of  $A$ .

$U^*AU = T$  implies  $A \approx T$ , and so  $\text{trace}(A) = \text{trace}(T)$  and  $\det(A) = \det(T)$ . Thus, we have

### Result

- $\text{trace}(A)$  is the sum of the eigenvalues of  $A$ .
- $\det(A)$  is the product of the eigenvalues of  $A$ .

# Normal Matrices

## Definition

A matrix  $A$  is said to be **normal** if  $AA^* = A^*A$ .

## Example

- *Real symmetric and Hermitian matrices are normal.*
- *A real Hermitian matrix is symmetric.*

## Result

- *If  $A$  is orthogonally diagonalizable then  $A$  is symmetric.*

*Note:*  $Q^t A Q = D \implies A = Q D Q^t \implies A^t = Q D^t Q^t = A$ .

- *If  $A$  is unitarily diagonalizable then  $A$  is normal.*

*Note:*  $A = U D U^* \implies$

$$\begin{aligned} A A^* &= (U D U^*)(U D^* U^*) \\ &= U D D^* U^* = U D^* D U^* \\ &= (U D^* U^*)(U D U^*) = A^* A. \end{aligned}$$

## Result

If  $A$  is a *Hermitian* or a *real symmetric* matrix, then all *eigenvalues* of  $A$  are *real*.

**Proof.** Let  $A^* = A$ ,  $A\mathbf{v} = \lambda\mathbf{v}$ ,  $\mathbf{v} \neq \mathbf{0}$ . Then

$$\begin{aligned}\lambda\|\mathbf{v}\|^2 &= \lambda(\mathbf{v} \cdot \mathbf{v}) = \mathbf{v} \cdot (\lambda\mathbf{v}) = \mathbf{v} \cdot (A\mathbf{v}) = \mathbf{v}^*(A\mathbf{v}) \\ &= (\mathbf{v}^*A)\mathbf{v} = (A\mathbf{v})^*\mathbf{v} = (\lambda\mathbf{v})^*\mathbf{v} = \bar{\lambda}\mathbf{v}^*\mathbf{v} = \bar{\lambda}\|\mathbf{v}\|^2.\end{aligned}$$

Therefore,  $\bar{\lambda} = \lambda$ , i.e.,  $\lambda \in \mathbb{R}$ . □

## Result

Let  $A$  be a *normal* matrix. Then for any  $\mathbf{x} \in \mathbb{C}^n$ ,  $\|A\mathbf{x}\| = \|A^*\mathbf{x}\|$ .

**Proof.** We have

$$\begin{aligned}\|A\mathbf{x}\|^2 &= (A\mathbf{x})^*(A\mathbf{x}) = \mathbf{x}^*(A^*A)\mathbf{x} = \mathbf{x}^*(AA^*)\mathbf{x} \\ &= (A^*\mathbf{x})^*(A^*\mathbf{x}) = \|A^*\mathbf{x}\|^2.\end{aligned}$$
□

## Result

Let  $A$  be a *normal* matrix, and  $\lambda$  be an eigenvalue of  $A$  with eigenvector  $\mathbf{u}$ . Then  $\bar{\lambda}$  is an eigenvalue of  $A^*$  with eigenvector  $\mathbf{u}$ .

**Proof.** Note that  $A - \lambda I$  is normal:

$$\begin{aligned}(A - \lambda I)(A - \lambda I)^* &= (A - \lambda I)(A^* - \bar{\lambda} I) \\&= AA^* - \bar{\lambda}A - \lambda A^* + |\lambda|^2 I \\&= A^*A - \bar{\lambda}A - \lambda A^* + |\lambda|^2 I \\&= (A - \lambda I)^*(A - \lambda I).\end{aligned}$$

Now

$$\begin{aligned}A\mathbf{u} = \lambda\mathbf{u} &\implies \|(A - \lambda I)\mathbf{u}\| = 0 \implies \|(A - \lambda I)^*\mathbf{u}\| = 0 \\&\implies \|(A^* - \bar{\lambda}I)\mathbf{u}\| = 0 \implies A^*\mathbf{u} = \bar{\lambda}\mathbf{u}.\end{aligned}$$



## Result

If  $A$  is a *normal* matrix, then *eigenvectors corresponding to distinct eigenvalues* of  $A$  are *orthogonal*.

**Proof.** Let  $A$  be normal. Let  $A\mathbf{u} = \lambda\mathbf{u}$ ,  $A\mathbf{v} = \mu\mathbf{v}$ ,  $\lambda \neq \mu$ . Then

$$\begin{aligned}\lambda(\mathbf{v}^*\mathbf{u}) &= \mathbf{v}^*(\lambda\mathbf{u}) = \mathbf{v}^*(A\mathbf{u}) = (\mathbf{v}^*A)\mathbf{u} = (A^*\mathbf{v})^*\mathbf{u} \\ &= (\bar{\mu}\mathbf{v})^*\mathbf{u} = \mu(\mathbf{v}^*\mathbf{u}).\end{aligned}$$

Therefore  $(\lambda - \mu)\mathbf{v}^*\mathbf{u} = 0$ , i.e.,  $\mathbf{v} \cdot \mathbf{u} = \mathbf{v}^*\mathbf{u} = 0$ . □

## Result

If  $A$  is a *Hermitain* or a *real symmetric* matrix, then *eigenvectors corresponding to distinct eigenvalues* of  $A$  are *orthogonal*.

## Result (Spectral Theorem for Normal Matrices)

A matrix  $A$  is *normal* iff there is an *unitary matrix*  $U$  such that  $U^*AU$  is a *diagonal matrix*.

**Proof.** Let  $U^*AU = D = \text{diag}(\lambda_1, \dots, \lambda_n)$ . Then  $A = UDU^*$ , and

$$\begin{aligned} A^*A &= (UDU^*)^*(UDU^*) = (UD^*U^*)(UDU^*) \\ &= UD^*DU^* = UDD^*U^* = AA^*, \end{aligned}$$

because  $D^*D = \text{diag}(|\lambda_1|^2, \dots, |\lambda_n|^2) = DD^*$ . Thus  $A$  is normal.

Next, let  $U$  be a unitary matrix such that  $U^*AU = T = [t_{ij}]$  is upper triangular (Schur Triangularization Theorem). Suppose  $A$  is normal. Then  $A^*A = AA^*$  implies

$$T^*T = (U^*AU)^*(U^*AU) = U^*A^*AU = U^*AA^*U = TT^*.$$

Now,  $(T^*T)_{11} = |t_{11}|^2$  and  $(TT^*)_{11} = \sum_{i=1}^n |t_{1i}|^2$ . Therefore,  $t_{12} = \dots = t_{1n} = 0$ . **Continue.** Conclude that  $t_{ij} = 0$  if  $i \neq j$ , i.e.,  $T$  is diagonal. □

## Result

A matrix  $A$  is *Hermitian* iff there is an *unitary matrix*  $U$  such that  $U^*AU$  is a *real diagonal* matrix.

**Proof.** Suppose  $A$  is Hermitian. Then  $A$  has only real eigenvalues. Also  $A$  is normal. So there is an *unitary matrix*  $U$  such that  $U^*AU$  is a *real* diagonal matrix.

Conversely, let  $U^*AU = D$ , where  $U$  is unitary and  $D$  is real diagonal. Then  $A^* = (UDU^*)^* = UD^*U^* = UDU^* = A$ , since  $D^* = D$ . □

## Result (Spectral Theorem for Real Symmetric Matrices)

A real matrix  $A$  is *symmetric* iff there is an *orthogonal matrix*  $Q$  such that  $Q^tAQ$  is a *diagonal* matrix.

**Hint.** If  $A$  is real symmetric, then its eigenvalues are real, so there are corresponding *real* eigenvectors. The eigenvectors are orthogonal.

## Result (Cayley-Hamilton Theorem)

Every square matrix  $A$  satisfies its characteristic equation  $p(x) = 0$ .

**Proof.** Let  $\lambda_1, \dots, \lambda_n$  be the eigenvalues of  $A$ . Then

$$p(x) = \prod_{i=1}^n (x - \lambda_i).$$

By **Schur's Triangularization Theorem**,  $U^*AU = T = [t_{ij}]$  for some unitary  $U$  and upper-triangular  $T$ . We have  $p(A) = Up(T)U^*$ , and so

$$p(A) = \mathbf{0} \iff p(T) = \mathbf{0} \iff (T - \lambda_1 I) \cdots (T - \lambda_n I) = \mathbf{0}.$$

Now,  $T - \lambda_1 I$  has the **first column zero**,  $(T - \lambda_1 I)(T - \lambda_2 I)$  has the first **two columns zero**, and so on. So,

$$(T - \lambda_1 I) \cdots (T - \lambda_n I) = \mathbf{0}.$$

Thus,  $p(A) = \mathbf{0}$ . □

## Result (Schur Unitary Triangularization Theorem)

For an  $n \times n$  matrix  $A$ , *there is* an *unitary matrix*  $U$  such that  $U^*AU$  is upper triangular.

**Proof.** By induction on  $n$ . True for  $n = 1$ . Assume the result be true for all  $(n - 1) \times (n - 1)$  matrices. Now let  $A$  be  $n \times n$ .

Let  $\lambda_1$  be an eigenvalue of  $A$ ,  $A\mathbf{w}_1 = \lambda_1\mathbf{w}_1$ , and  $\|\mathbf{w}_1\| = 1$ . Let  $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$  be an orthonormal basis of  $\mathbb{C}^n$ . Put  $W = [\mathbf{w}_1 \cdots \mathbf{w}_n]$ . Then

$$W^*AW = \begin{bmatrix} \mathbf{w}_1^* \\ \vdots \\ \mathbf{w}_n^* \end{bmatrix} [A\mathbf{w}_1 \cdots A\mathbf{w}_n] = \begin{bmatrix} \lambda_1 & * \\ 0 & A' \end{bmatrix},$$

because  $A\mathbf{w}_1 = \lambda_1\mathbf{w}_1$  and  $\mathbf{w}_j^*\mathbf{w}_1 = 0$ ,  $2 \leq j \leq n$ . Since  $A'$  is  $(n - 1) \times (n - 1)$ , by induction hypothesis, we get a unitary matrix  $U'$  such that  $U'^*A'U' = T'$  is upper-triangular.

Take  $U = W \begin{bmatrix} 1 & 0 \\ 0 & U' \end{bmatrix}$ . Then  $U$  is unitary:

$$U^* U = \left( \begin{bmatrix} 1 & 0 \\ 0 & U'^* \end{bmatrix} W^* \right) \left( W \begin{bmatrix} 1 & 0 \\ 0 & U' \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 \\ 0 & I_{n-1} \end{bmatrix} = I_n.$$

We now have

$$\begin{aligned} U^* A U &= \begin{bmatrix} 1 & 0 \\ 0 & U'^* \end{bmatrix} W^* A W \begin{bmatrix} 1 & 0 \\ 0 & U' \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & U'^* \end{bmatrix} \begin{bmatrix} \lambda_1 & * \\ 0 & A' \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & U' \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 & * \\ 0 & U'^* A' U' \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 & * \\ 0 & T' \end{bmatrix}, \text{ an upper-triangular matrix.} \end{aligned}$$

Hence, we conclude by principle of mathematical induction that the result is true for all  $n$ . □

Thank you.

Wish you all the best.